

Name: _____

Date: _____

DIVISION, (BY 10, 100 AND 1000)

LO: I can divide whole numbers and decimals by 10, 100 and 1000 by shifting place value.

How does $\div 10$ work?

Dividing by **10** shifts every digit one column **RIGHT**. The digits stay the same; their place value gets smaller.

Common error: 'just remove a zero.' Doesn't work for decimals or numbers without trailing zeros (e.g. $47 \div 10 \neq 47$, it's 4.7).

Examples of place-value shift

●	$70 \div 10 = 7$	shift R by 1
●	$700 \div 100 = 7$	shift R by 2
●	$7000 \div 1000 = 7$	shift R by 3
●	$47 \div 10 = 4.7$	decimal appears
●	$47 \div 10 = 47$	must shift, not erase

Key idea

Dividing by 10 = shift digits 1 right. By 100 = shift 2 right. By 1000 = shift 3 right. The decimal point appears to move LEFT.

$$\textcircled{345} \div \textcircled{100} = \textcircled{3.45}$$

Decimal point moves 2 places left

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - whole $\div 10$

Calculate $80 \div 10$

80 has digits 8 and 0
Shift each one column right
= 8

= 8

The 8 moves from tens to ones; the 0 falls off.

Example 2 - whole $\div 100$

Calculate $4500 \div 100$

Each digit shifts 2 places right
 $4500 \rightarrow 45.00$
= 45

= 45

Two zeros become placeholders that drop off.

Example 3 - needs decimal

Calculate $23 \div 10$

The 3 moves to tenths
Decimal point appears
= 2.3

= 2.3

We can't just 'remove a zero' here — there isn't one.

Example 4 - decimal $\div 1000$

Calculate $47 \div 1000$

Shift each digit 3 places right
 $47 \rightarrow 0.047$
= 0.047

= 0.047

Add zeros as placeholders before the digits.

